# 数理方程

第四周

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1) 多自变量 2) 非齐次PDE

# 多自变量 $n>2$ 齐次方程

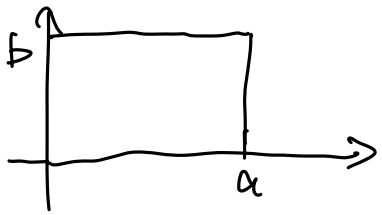
PDE $\longrightarrow$ ODE + ODE

本征值

PDE $\longrightarrow$ ODE + ODE + ODE

本征值

二维扩散




$$u_t = Dv^2u = D\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$0<x<a,\quad 0<y<b$

$$u|_{t=0}=\phi(x,y),\quad \frac{\partial u}{\partial x}\Big|_{x=0}=\frac{\partial u}{\partial x}\Big|_{x=a}=\frac{\partial u}{\partial y}\Big|_{y=0}=\frac{\partial u}{\partial y}\Big|_{y=b}=0$$

1) 分离变量 $u(x,y,t)=X(x)Y(y)T(t)$

$$PDE\Rightarrow \frac{1}{D}XYT'=X''YT+XY''T\Rightarrow \frac{T'}{DT}=\frac{X''}{X}+\frac{Y''}{Y}=-\lambda$$

$$-\mu=\frac{X''}{X}=-\lambda-\frac{Y''}{Y}$$

$x$ 的函数　　$y$ 的函数

$=-\nu$

$$\Rightarrow \frac{X''}{X}=-\mu,\quad \frac{Y''}{Y}=-\nu,\quad \lambda=\mu+\nu$$

$$\frac{\partial u}{\partial x}\Big|_{x=0}=X'(0)Y(y)T(t)=0\Rightarrow X'(0)=0 \qquad X''+\mu X=0$$

$$X'(a)=0$$

$$Y'(0)=0 \qquad Y''+\nu Y=0$$

$$Y'(b)=0$$

2) $\mu=0$: $X_0=Ax+B\Rightarrow X_0(x)=1$

$\mu>0$: ~~$A\sin\sqrt{\mu}x$~~ $+B\cos\sqrt{\mu}x\Rightarrow X_n(x)=\cos\frac{n\pi x}{a}$, $n=1,2,3\cdots$

$\mu<0$: 无解

$X'(0)=\sqrt{\mu}A=0$

$X'(a)=0\Rightarrow\sqrt{\mu}=\frac{n\pi}{a}$

$\lambda_n=\left(\frac{n\pi}{a}\right)^2$

$$\mu_n=\left(\frac{n\pi}{a}\right)^2,\quad n=0,1,2\cdots \qquad X_n(x)=\cos\frac{n\pi x}{a}$$

$y$: $\nu_m=\left(\frac{m\pi}{b}\right)^2$, $m=0,1,2\cdots$ $\quad Y_m(y)=\cos\frac{m\pi y}{b}$

$$\lambda_{nm}=\mu_n+\nu_m=\left(\frac{n\pi}{a}\right)^2+\left(\frac{m\pi}{b}\right)^2$$

3) $T_{nm}'(t) = -\lambda_{nm} D T_{nm}(t) \Rightarrow T_{nm} = A_{nm} e^{-\lambda_{nm} D t}$

↑常数

$u_{nm}(x,y,t) = X_n Y_m T_{nm}$

4) $u(x,y,t) = \sum_{n,m} u_{nm}(x,y,t) = \sum_{n,m=0}^{\infty} A_{nm} \cos\frac{n\pi x}{a} \cos\frac{m\pi y}{b} e^{-D[(\frac{n\pi}{a})^2+(\frac{m\pi}{b})^2]t}$

↑二重傅里叶级数

$u|_{t=0} = \phi(x,y) = \sum_{n,m=0}^{\infty} A_{nm} \underbrace{\cos\frac{n\pi x}{a}}_{X_n} \underbrace{\cos\frac{m\pi y}{b}}_{Y_m}$

正交性 $\int_0^a X_n X_{n'} dx = \begin{cases} \frac{a}{2}, & n=n'\neq 0 \\ a, & n=n'=0 \\ 0 & n\neq n' \end{cases} = \frac{a}{2}(1+\delta_{n0})\delta_{nn'}$

↑ $\delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$\int_0^b Y_m Y_{m'} dy = \frac{b}{2}(1+\delta_{m0})\delta_{mm'}$

$\int_0^a \int_0^b \phi(x,y) \cos\frac{n\pi x}{a} \cos\frac{m\pi y}{b} dx dy = \underbrace{\frac{a}{2}(1+\delta_{n0})\frac{b}{2}(1+\delta_{m0}) A_{nm}}$

$A_{nm} = \frac{4}{ab} \frac{1}{(1+\delta_{n0})(1+\delta_{m0})} \int_0^a \int_0^b \phi(x,y) \cos\frac{n\pi x}{a} \cos\frac{m\pi y}{b} dx dy$

$\{X_n(x) Y_m(y) \mid n,m = 0,1,2\ldots\}$ 正交基

## 非齐次方程，三种解法




$$u_{tt}-a^2u_{xx}=f(x,t) \quad 0<x<l,\ t>0$$

$$u|_{x=0}=u|_{x=l}=0 \qquad u|_{t=0}=\frac{\partial u}{\partial t}\Big|_{t=0}=0$$

1) 齐次化原理. $w(x,t;\tau)$,

$$\begin{cases} w_{tt}-a^2w_{xx}=0, & 0<x<l,\ t>\tau \\ w|_{t=\tau}=0, \quad \dfrac{\partial w}{\partial t}\Big|_{t=\tau}=f(x,\tau) \end{cases}$$

$$\Rightarrow u(x,t)=\int_0^t w(x,t;\tau)\,d\tau, \qquad w|_{x=0}=w|_{x=l}=0$$

$$w(x,t;\tau)=\sum_{n=1}^{\infty} C_n(\tau)\sin\frac{n\pi a}{l}(t-\tau)\sin\frac{n\pi x}{l}$$

$$C_n=\frac{2}{n\pi a}\int_0^l f(x,\tau)\sin\frac{n\pi x}{l}\,dx$$

$$=\frac{l}{n\pi a}\left[\frac{2}{l}\int_0^l f(x,\tau)\sin\frac{n\pi x}{l}\right]=\frac{l}{n\pi a}f_n(\tau)$$

$=f_n$, $f$ 的第 $n$ 个 Fourier 系数

初位移中=0，初速度中=f

回忆齐次PDE：$C_n=\frac{2}{n\pi a}\int_0^l \psi(x)\sin\frac{n\pi x}{l}dx$

$$\Rightarrow u(x,t)=\int_0^t w(x,t;\tau)d\tau=\sum_{n=1}^{\infty}\left[\int_0^t C_n(\tau)\sin\frac{n\pi a}{l}(t-\tau)d\tau\right]\sin\frac{n\pi x}{l}$$

$$=\sum_{n=1}^{\infty}\left[\int_0^t \frac{l}{n\pi a}f_n(\tau)\sin\frac{n\pi a}{l}(t-\tau)d\tau\right]\sin\frac{n\pi x}{l}$$

$=T_n(t)$ $\qquad X_n(x)$

$$=\sum_{n=1}^{\infty}T_n(t)X_n(x)$$

例：$u_{tt}-a^2u_{xx}=f(x,t)=g(x)\sin\omega t$, $\quad 0<x<l,\ t>0$

$$u|_{x=0}=u|_{x=l}=0 \qquad u|_{t=0}=\frac{\partial u}{\partial t}\Big|_{t=0}=0$$

$$\Rightarrow T_n(t)=\frac{l}{n\pi a}\int_0^t f_n(\tau)\sin\frac{n\pi a}{l}(t-\tau)d\tau=\frac{1}{\omega_n}\int_0^t f_n(\tau)\sin\omega_n(t-\tau)d\tau$$

记为 $\frac{1}{\omega_n}$ $\qquad \omega_n$

$$=\frac{1}{\omega_n}\int_0^t g_n(\tau)\sin\omega\tau\sin\omega_n(t-\tau)d\tau$$

↑ $g(x)$ 的傅里叶系数

若 $g(x)=A$（常数）$\Rightarrow g_n=\frac{2A}{\pi}\frac{1-(-1)^n}{n}$

$f_n(t)=g_n(t)\cos\omega t$

$\omega\neq\omega_n$ 时 $=\frac{g_n}{\omega_n}\frac{\omega\sin\omega_n t-\omega_n\sin\omega t}{\omega^2-\omega_n^2}$

$$\Rightarrow u(x,t)=\sum_{n=1}^{\infty}X_n(x)T_n(t)=\sum_{n=1}^{\infty}\frac{g_n}{\omega_n}\frac{\omega\sin\omega_n t-\omega_n\sin\omega t}{\omega^2-\omega_n^2}\sin\frac{n\pi x}{l}$$

$\omega\to\omega_k=\frac{k\pi a}{l}\Rightarrow$ 分母 $\omega^2-\omega_k^2\to 0$

当 $\omega=\omega_k$ 时，用洛必达法则：分子分母对 $\omega$ 求导后取 $\omega=\omega_k$

↓也可不用此法，直接从原始积分得到

$$\Rightarrow u(x,t)=\frac{g_k}{2\omega_k^2}(\sin\omega_k t-\omega_k t\cos\omega_k t)\sin\frac{k\pi x}{l}+\sum_{k\neq n}(\cdots)\sin\frac{n\pi x}{l}$$

2) 本征函数展开 $u(x,t)=\sum_{n=1}^{\infty}T_n(t)X_n(x)$

$T_n(t)$：待求；$X_n(x)$：由相应的齐次PDE加同样的齐次BC得出。

$$f(x,t)=\sum_{n=1}^{\infty}f_n(t)X_n(x)$$

$$u_{tt}-a^2u_{xx}=f(x,t)\Rightarrow\sum_{n=1}^{\infty}T_n''X_n-a^2\sum_{n=1}^{\infty}T_nX_n''=\sum_{n=1}^{\infty}f_nX_n$$

（假定级数可逐项求导）

$X_n''=-\lambda_nX_n$

$$\Rightarrow\sum_{n=1}^{\infty}(T_n''+a^2\lambda_nT_n-f_n)X_n=0$$

$$\Rightarrow T_n''(t)+a^2\lambda_nT_n(t)=f_n(t),\quad \lambda_n=\left(\frac{n\pi}{l}\right)^2$$

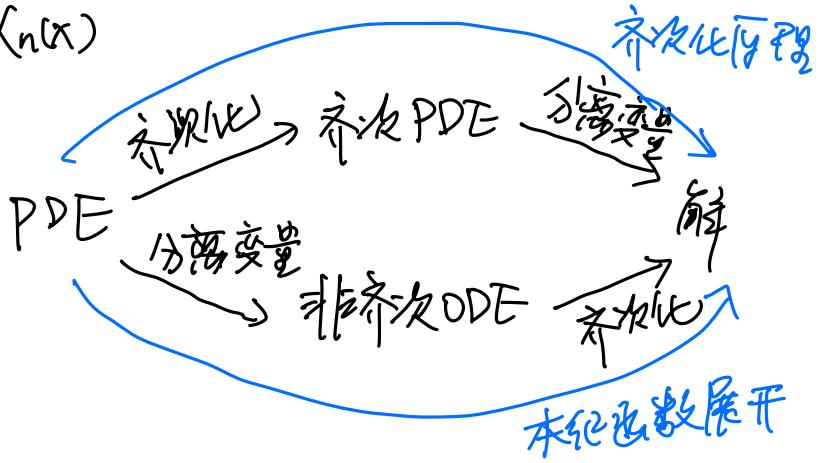
$$T_n''+\left(\frac{n\pi a}{l}\right)^2T_n=f_n(t)$$

$$T_n(t)=\int_0^t w_n(t;\tau)\,d\tau$$

$\Leftarrow$ $w_n''+\left(\frac{n\pi a}{l}\right)^2w_n=0$；$w_n(t;\tau)\big|_{t=\tau}=0$，$w_n'(t;\tau)\big|_{t=\tau}=f_n(\tau)$

$$=\int_0^t\frac{l}{n\pi a}\sin\frac{n\pi a}{l}(t-\tau)f_n(\tau)\,d\tau$$

$\Rightarrow u(x,t)=\sum_{n=1}^{\infty} T_n(t)X_n(x)$




3) 特解法

例 (不涉及分离变量，但思路一样)

$$u_{tt}-a^2u_{xx}=\sin(Ax+Bt) \qquad -\infty<x<\infty,\ t>0$$

$$u|_{t=0}=\frac{\partial u}{\partial t}\Big|_{t=0}=0$$

$\Rightarrow v(x,t)$, $v_{tt}-a^2v_{xx}=\sin(Ax+Bt)$,

$v=C\sin(Ax+Bt) \Rightarrow C(-B^2+A^2a^2)\sin(\cdots)=\sin(\cdots)$

$$\Rightarrow v=\frac{\sin(Ax+Bt)}{A^2a^2-B^2}$$

但 $v$ 不满足初始条件

$u=v+w$ ← 满足 $w_{tt}-a^2w_{xx}=0$ ← 齐次PDE

$$w|_{t=0}=-v|_{t=0},\quad \frac{\partial w}{\partial t}\Big|_{t=0}=-\frac{\partial v}{\partial t}\Big|_{t=0}$$

解出 $w$，$v+w=u$ 即解.

例2: $u_{tt} - a^2 u_{xx} = f(x)$ ← 不依赖于 $t$ $\quad 0<x<l,\ t>0$

$u|_{x=0} = u|_{x=l} = 0, \quad u|_{t=0} = u_t|_{t=0} = 0$

取特解 $v(x,t) = v(x)$ 满足 PDE 与 BC:

$-a^2 v_{xx} = f(x) \Rightarrow v_{xx} = -\frac{1}{a^2} f(x)$

$v(0) = v(l) = 0$ 解此 ODE 得 $v(x)$.

例如 $f(x) = Ax \Rightarrow$ 由 ODE 得 $v(x) = -\frac{A}{a^2}\left(\frac{x^3}{6} + Cx + D\right)$

$= \frac{A}{6a^2}(l^2 x - x^3)$

由 $v(0) = v(l) = 0$ 定 $C, D$

$v$ 不满足 IC.

$u = v + w$ ($w$ 待求) $\xrightarrow{\text{代入 PDE}}$ $w_{tt} - a^2 w_{xx} = -(v_{tt} - a^2 v_{xx}) + f(x) = 0$

齐次 PDE

齐次 BC:
$$\begin{cases} w|_{x=0} = u|_{x=0} - v|_{x=0} = 0, \\ w|_{x=l} = u|_{x=l} - v|_{x=l} = 0. \end{cases}$$

IC:
$$\begin{cases} w|_{t=0} = u|_{t=0} - v|_{t=0} \\ w_t|_{t=0} = u_t|_{t=0} - v_t|_{t=0} \end{cases}$$

解出 $w$. $\quad v + w = u$ 即解.

# 扩散方程 $u_t = D u_{xx} + f(x,t)$, $0<x<l$, $t>0$, $u|_{t=0}=0$

(对t)所PDE的齐次化原理

L线性,且不含对t的各阶偏导 $u_t, u_{tt}, \cdots$

$$\frac{\partial u}{\partial t} = L(u) + f(\vec{x},t),\ t>0 \qquad u|_{t=0}=0$$

$$\Longleftarrow \begin{cases} \frac{\partial w}{\partial t} = L(w),\ t>\tau \\ w(\vec{x},t;\tau)|_{t=\tau} = f(x,\tau) \end{cases}$$

$$u(x,t) = \int_0^t w(\vec{x},t;\tau)\,d\tau$$

验证: $u|_{t=0}=0$ 显然.

$$\frac{\partial u}{\partial t} = w(\vec{x},t;\tau)|_{t=\tau} + D\int_0^t \frac{\partial w}{\partial t}\,d\tau$$

$$= f(x,t) + D\int_0^t \frac{\partial^2 w}{\partial x^2}\,d\tau$$

$$= f(x,t) + D\frac{\partial^2}{\partial x^2}\left[\int_0^t w(\vec{x},t;\tau)\,d\tau\right] = f(x,t) + D u_{xx}$$

理解:将时间分割为小区间,对所有小区间里源f的贡献积分

(与波动方程思路类似)

调用齐次PDE结果:

$$w_t - D w_{xx} = 0 \quad 0<x<l,\ t>0$$

$$\begin{cases} w(x,t;\tau)|_{t=\tau} = f(x,\tau) \end{cases}$$

$$w(x,t;\tau) = \sum_{n=1}^{\infty} f_n(\tau)\, e^{-D(\frac{n\pi}{l})^2 (t-\tau)} \sin\frac{n\pi x}{l}$$

$$f_n(\tau) = \frac{2}{l}\int_0^l f(x,\tau)\sin\frac{n\pi x}{l}\,dx$$

$$\Rightarrow u(x,t)=\int_0^t w(x,t;\tau)d\tau=\sum_{n=1}^{\infty}\underbrace{\left[\int_0^t f_n(\tau)e^{-D(\frac{n\pi}{l})^2(t-\tau)}d\tau\right]}_{=T_n(t)}\underbrace{\sin\frac{n\pi x}{l}}_{=X_n(x)}$$

$$=\sum_{n=1}^{\infty}T_n(t)X_n(x)$$

同样结果

也可本征函数展开法：$\underbrace{T_n(t)}_{\text{待求}}X_n(x)$

# 高阶齐次化原理 （不作要求）

若 $w(\vec{x}, t; \tau)$ 满足

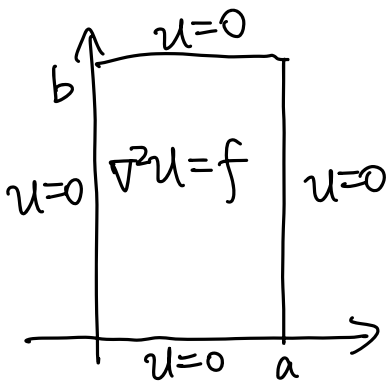
$L$ 线性，其中关于 $t$ 的最高阶偏导数不超过 $m-1$ 阶

$$\begin{cases} \dfrac{\partial^m w}{\partial t^m} = L(w), \quad t > \tau \\ w\big|_{t=\tau} = \dfrac{\partial w}{\partial t}\Big|_{t=\tau} = \cdots = \dfrac{\partial^{m-2} w}{\partial t^{m-2}}\Big|_{t=\tau} = 0 \\ \dfrac{\partial^{m-1} w}{\partial t^{m-1}}\Big|_{t=\tau} = f(x, \tau) \end{cases}$$

则 $u(\vec{x}, t) = \int_0^t w(\vec{x}, t; \tau)\, d\tau$ 满足

$$\begin{cases} \dfrac{\partial^m u}{\partial t^m} = L(u) + f(\vec{x}, t), \quad t > 0 \\ u\big|_{t=0} = \dfrac{\partial u}{\partial t}\Big|_{t=0} = \cdots = \dfrac{\partial^{m-1} u}{\partial t^{m-1}}\Big|_{t=0} = 0 \end{cases}$$

# 泊松方程




$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=f(x,y)$$

$$0<x<a,\quad 0<y<b$$

$$u\big|_{x=0}=u\big|_{x=a}=u\big|_{y=0}=u\big|_{y=b}=0$$

解法一：本征函数展开法

$$u(x,y)=\sum_{n=1}^{\infty} X_n(x)\,Y_n(y)$$

$X_n(x)$：本征函数 $X_n=\sin\frac{n\pi x}{a}$；$Y_n(y)$：待求

泊松 $(x,y)$ ⟷ 波动 $(x,t)$

$X_n(x)Y_n(y)$ ⟷ $X_n(x)T_n(t)$

$Y_n(y)$：待求；$T_n(t)$：待求

解法二：二重级数展开

$$u(x,y)=\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} C_{nm}\,X_n(x)\,X_m(y)$$

$C_{nm}$：待求系数

$$f(x,y)=\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} f_{nm}\,X_n(x)\,X_m(y)$$

代入 PDE $\Rightarrow$ $C_{nm}\left[-\left(\frac{n\pi}{a}\right)^2-\left(\frac{m\pi}{b}\right)^2\right]=f_{nm}\Rightarrow C_{nm}=\cdots\cdots$

此解法无须解ODE，代价为级数多一重，和一重级数相比，一般需要保留更多项才能达到同样精度.

解法三：特解法

根据 $f(x,y)$ 的具体形式选取特解 $v(x,y)$ 满足

$$\nabla^2 v = f,$$

$$u = v + w, \quad \nabla^2 w = 0.$$

求解 $w$

逐次求导的危险性

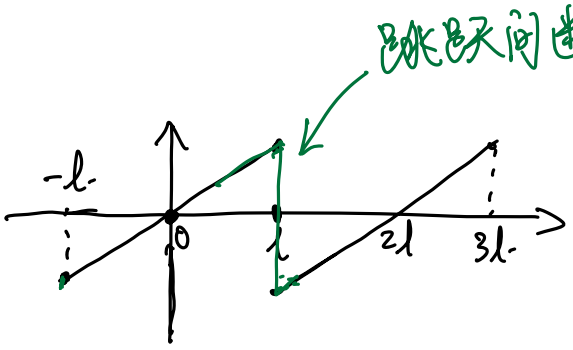
$$f(x) = x = \sum_{n=1}^{\infty} \frac{2l}{n\pi}(-1)^{n+1}\sin\frac{n\pi x}{l}$$

⇓ 求导

$$1 \neq \sum_{n=1}^{\infty} 2(-1)^{n+1}\cos\frac{n\pi x}{l}$$

⇓ 求导

$$0 \neq \cdots$$




$$f(x) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{l}$$

存在跳跃间断点 $\Rightarrow B_n \sim \frac{1}{n}$

包括区间端点处的跳跃



$\phi(x)$有跳跃：$B_n \sim \frac{1}{n}$

$\phi$连续，$\phi'$跳跃 $B_n \sim \frac{1}{n^2}$

$\phi, \phi'$连续，$\phi''$跳跃：$B_n \sim \frac{1}{n^3}$

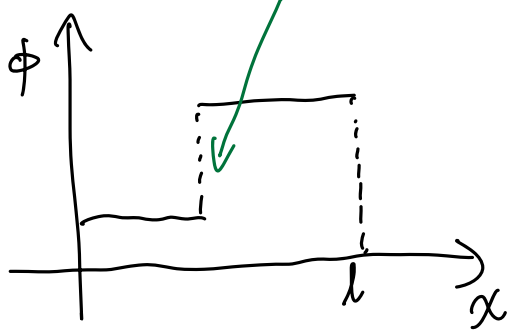
$\phi, \cdots \phi^{(k-1)}$连续，$\phi^{(k)}$跳跃：$B_n \sim \frac{1}{n^{k+1}}$

$$\phi(x) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{l}$$

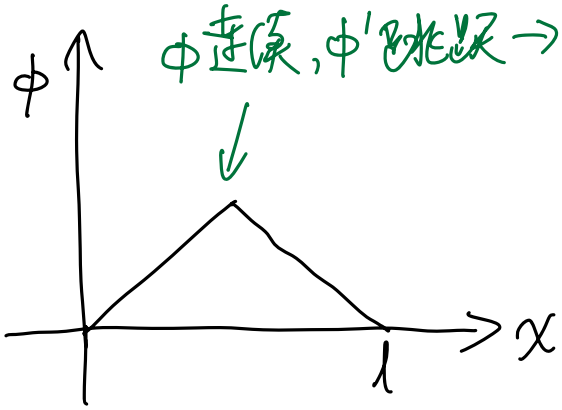
$$B_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi x}{l}dx$$

$B_n$衰减越慢，逐项求导越不易收敛

$\phi(x)$跳跃(不连续)$\to B_n \sim \frac{1}{n}$

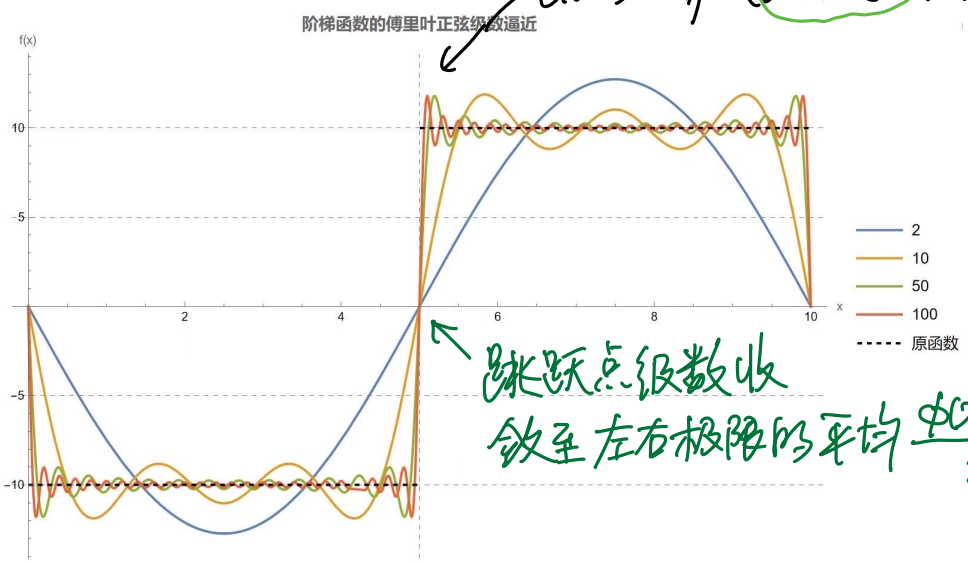



$\phi$连续，$\phi'$跳跃$\to B_n \sim \frac{1}{n^2}$




$\phi(x-0) \neq \phi(x+0)$

Gibbs现象(跳跃点附近振荡)$\Rightarrow$收敛但非一致收敛




跳跃点级数收敛至左右极限的平均 $\frac{\phi(x+0)+\phi(x-0)}{2}$

(不作要求)